

Colt Melton

Kimball, MI • (810) 479-6235 • coltmeltonwork@gmail.com

[linkedin.com/in/coltmelton](https://www.linkedin.com/in/coltmelton) • github.com/coltmelton

PROFESSIONAL SUMMARY

Computer Science student at Kettering University with professional experience developing software, automation frameworks, cloud-integrated applications, and machine learning systems in industrial engineering environments. Experienced in C#, Python, AWS, Linux, REST APIs, and distributed software development. Proven ability to design, deploy, and maintain production software solutions while collaborating across multidisciplinary engineering teams. Interested in defense technology, autonomous systems, infrastructure software, and mission-critical applications.

EDUCATION

Kettering University | Flint, MI

Bachelor of Science, Computer Science

Dean's List | GPA: 3.5+

July 2023 – Present

Ajou University | Suwon, South Korea

Exchange Program in Software Engineering

February 2025 – July 2025

TECHNICAL SKILLS

Languages: C#, Python, SQL, C++, Java, JavaScript, C

Software & Development: .NET, REST APIs, GUI Development, Git, Bamboo, Machine Learning, Data Analysis

Infrastructure & Cloud: AWS, Linux, Networking, System Administration

Cybersecurity: SIEM Development, Vulnerability Assessment, Incident Response, Network Security

PROFESSIONAL EXPERIENCE

Endress+Hauser | Ann Arbor, MI

Research & Development Engineer / Scientist

October 2025 – October 2026

- Designed and deployed a customer facing spectroscopy visualization dashboard intended for customers to assess analyzer performance, spectral quality, and system behavior in real time.
- Supported deployment automation and validation pipelines utilizing Bamboo.
- Collaborated with scientists and engineers to improve instrumentation reliability and analytical capabilities.

Bosch International Engineering | Farmington Hills, MI

Calibration Engineer, Braking Systems & Chassis

October 2023 – July 2025

- Developed automation tools utilizing proprietary engineering APIs to improve testing and engineering workflows.
- Assisted in calibration, validation, and performance analysis of prototype braking systems.
- Collaborated with multidisciplinary engineering teams supporting advanced vehicle chassis technologies and safety systems.

RESEARCH EXPERIENCE

Senior Thesis: Machine Learning, Neural Network Raman Spectral Decomposition

- Led the design and development of a machine-learning-guided Raman spectral decomposition pipeline
- Directed research, development, validation, and presentation of a ML analysis pipeline.
- Authored and presented a comprehensive technical thesis

PROJECTS

VMware Automation Platform

- Developed a C# automation framework for VM deployment, validation, and software testing workflows.

SCIF SIEM

- Developed a Python Security Information and Event Management platform utilizing custom detection rules, event correlation, and MITRE ATT&CK mapping.

OPC UA Spectroscopy Dashboard

- Developed and deployed a real-time industrial monitoring dashboard integrating live spectroscopy data streams for analyzer visualization and performance assessment.

LEADERSHIP & ACTIVITIES

- Kagle Leadership Initiative(KLI) - Support and guidance for leadership and academics amongst young college and high school students
- President, Cliff Hangers Rock Climbing Club - Scheduling, planning, event management, budget

Activities: Hackathons • Ethical Hacking • Boxing • Backpacking • Weightlifting • Guitar

CERTIFICATIONS

CompTIA IT Fundamentals (ITF+)